

A common inverse estimate from row Cramer

MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound

theorem

Controls all near-maximal inverse frames by an explicit basis-dependent constant.

Statement

Let $n \in \mathbb{N}$, let E be a finite-dimensional real normed vector space, and fix an ordered basis $b = (b_j)_{j=1}^n$. Write $C_b : E \rightarrow \mathbb{R}^n$ for the coordinate isomorphism, defined by $x = \sum_{j=1}^n (C_b x)_j b_j$, and give its coordinate space the sup norm $\|c\|_\infty = \max_j |c_j|$ (zero in dimension zero). A frame is an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals, with no independence assumed. Its matrix and evaluation map are

$$M_F = (\ell_i(b_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F C_b x.$$

The admissible frames form the product of closed dual unit balls

$$\mathcal{U} = \{F : \|\ell_i\|_{E^*} \leq 1 \text{ for every } i\}.$$

The norm on E need not be a coordinate sup norm. Let $e_i^*(x) = (C_b x)_i$ be the dual coordinate functionals, and put $D_b = \max_{F \in \mathcal{U}} |\det M_F|$ and $\alpha_b = (1 + \sum_i \|e_i^*\|_{E^*})^{-1}$. Fix $\eta < D_b$. Define

$$A_b(\eta) = \frac{n D_b}{\alpha_b (D_b - \eta)}, \quad K_b(\eta) = 1 + \|C_b^{-1}\|_{\text{op}} A_b(\eta).$$

For $F \in \mathcal{N}_{b,\eta} = \{F \in \mathcal{U} : D_b - \eta \leq |\det M_F|\}$ and every $c \in \mathbb{R}^n$,

$$\|C_b^{-1}(M_F^{-1}c)\|_E \leq K_b(\eta) \|c\|_\infty.$$

Assumptions

The only deficit hypothesis is $\eta < D_b$; the frame satisfies both the dual-row contraction condition and the displayed determinant lower bound. The coefficient c is arbitrary. There is no extra $\eta \geq 0$ hypothesis.

Conclusion

The same positive $K_b(\eta)$ works for every frame in this near-maximal set and every coefficient. The coordinate inverse C_b^{-1} and matrix inverse M_F^{-1} are distinct maps; the resulting norm is the given norm of E .

Proof route

Use row Cramer with a scaled coordinate functional, divide by the determinant gap, and transport the coordinate bound through the inverse coordinate map.

▼ Proof steps

1. **Read the row-Cramer identity one coordinate at a time.** Since $F \in \mathcal{N}_{b,\eta}$ and $\eta < D_b$,

$$|\det M_F| \geq D_b - \eta > 0.$$

For $r \in E^*$, let $M_F[i \leftarrow r]$ denote the matrix obtained by replacing row i with $(r(b_k))_{k=1}^n$. The row-Cramer formula is

$$\det M_F[i \leftarrow r] = (\det M_F) \sum_{k=1}^n r(b_k)(M_F^{-1})_{ki}.$$

Here i specifies the replaced row, while k runs over the basis coordinates. Apply provider row form of Cramer to M_F and the row $(r(b_k))_k$; its determinant hypothesis is the nonvanishing just shown. The provider uses $M_F^\top$, so its column replacement is exactly the row replacement above.

2. **Multiply by the coefficients and show the two sums.** For the same functional r and the given column c ,

$$\begin{aligned} \sum_{i=1}^n c_i \det M_F[i \leftarrow r] &= (\det M_F) \sum_{i=1}^n \sum_{k=1}^n c_i r(b_k)(M_F^{-1})_{ki} \\ &= (\det M_F) \sum_{k=1}^n r(b_k) \left(\sum_{i=1}^n (M_F^{-1})_{ki} c_i \right) \\ &= (\det M_F) \sum_{k=1}^n r(b_k)(M_F^{-1}c)_k. \end{aligned}$$

The second line interchanges two finite sums; the third is matrix multiplication. Finally, linearity of r and the definition of C_b^{-1} give

$$\begin{aligned} \sum_{k=1}^n r(b_k)(M_F^{-1}c)_k &= r \left(\sum_{k=1}^n (M_F^{-1}c)_k b_k \right) \\ &= r(C_b^{-1}(M_F^{-1}c)). \end{aligned}$$

Thus these calculations give the row-replacement identity in basis coordinates, with this same F, r, c . The column $M_F^{-1}c$ is a coordinate vector; only $C_b^{-1}(M_F^{-1}c)$ is the corresponding vector of E .

3. **Select one coordinate and bound its numerator.** Fix j and take $r = \alpha_b e_j^*$. The definition of α_b gives

$$\|\alpha_b e_j^*\|_{E^*} = \frac{\|e_j^*\|_{E^*}}{1 + \sum_k \|e_k^*\|_{E^*}} \leq 1.$$

This is scaled-row admissibility. Hence replacing any row i of the admissible frame F with this r gives another admissible frame, by admissible row replacement, and the determinant maximum gives

$$|\det M_F[i \leftarrow \alpha_b e_j^*]| \leq D_b \quad (1 \leq i \leq n).$$

In Step 2 the functional evaluation is now exactly

$$(\alpha_b e_j^*)(C_b^{-1}(M_F^{-1}c)) = \alpha_b (M_F^{-1}c)_j.$$

Taking absolute values in that identity and bounding each summand yields

$$\begin{aligned}\alpha_b |\det M_F| |(M_F^{-1}c)_j| &= \left| \sum_i c_i \det M_F[i \leftarrow \alpha_b e_j^*] \right| \\ &\leq \sum_i |c_i| |\det M_F[i \leftarrow \alpha_b e_j^*]| \\ &\leq \sum_i \|c\|_\infty D_b = n D_b \|c\|_\infty.\end{aligned}$$

This is the Cramer numerator bound for the same frame, coefficient column and scaled coordinate functional.

4. Divide by the gap and return to the given norm. Since $\alpha_b > 0$ and $|\det M_F| \geq D_b - \eta > 0$, the previous estimate implies

$$\begin{aligned}\alpha_b (D_b - \eta) |(M_F^{-1}c)_j| &\leq n D_b \|c\|_\infty, \\ |(M_F^{-1}c)_j| &\leq \frac{n D_b}{\alpha_b (D_b - \eta)} \|c\|_\infty = A_b(\eta) \|c\|_\infty.\end{aligned}$$

This is the coordinatewise inverse estimate. Taking the maximum over j and then applying the operator norm of C_b^{-1} gives

$$\begin{aligned}\|M_F^{-1}c\|_\infty &\leq A_b(\eta) \|c\|_\infty, \\ \|C_b^{-1}(M_F^{-1}c)\|_E &\leq \|C_b^{-1}\|_{\text{op}} \|M_F^{-1}c\|_\infty \\ &\leq \|C_b^{-1}\|_{\text{op}} A_b(\eta) \|c\|_\infty \\ &\leq K_b(\eta) \|c\|_\infty.\end{aligned}$$

The argument with a coordinate j is for $n > 0$. When $n = 0$, both vectors are zero, $A_b(\eta) = 0$ and $K_b(\eta) = 1$, so the conclusion is immediate. In every dimension $A_b(\eta) \geq 0$ and the leading 1 in $K_b(\eta)$ ensures positivity.

The coordinate factor and ambient inverse constant are the displayed A_b, K_b ; positivity of that constant requires the same gap. the common inverse-bound structure stores exactly a real K , a proof $K > 0$, and the uniform inverse estimate. the nonzero determinant criterion and reconstruction through the inverse frame also give $C_b^{-1}M_F^{-1}T_Fx = x$, so T_F is injective, as in frame-map injectivity. The functional row is evaluation on the basis; expansion of a functional in these coordinates and the transposed Cramer replacement justify the scalar identity used above. The row replacement operation and replacement determinant keep the frame, its matrix and its evaluation map distinct.

Main citations

- provider row form of Cramer — `Matrix.det_smul_inv_vecMul_eq_cramer_transpose`
- the row-replacement identity in basis coordinates — `MathlibAnnex.DeterminantFrame.sum_mul_replacementDeterminant`
- scaled-row admissibility — `MathlibAnnex.DeterminantFrame.coordinateFrame_mem`
- admissible row replacement — `MathlibAnnex.DeterminantFrame.replaceRow_mem_unitFrameSet`
- the maximum bound on every replaced determinant — `MathlibAnnex.DeterminantFrame.abs_replacementDeterminant_le_maximum`
- the Cramer numerator bound — `MathlibAnnex.DeterminantFrame.abs_cramerNumerator_le`
- coordinatewise inverse estimate — `MathlibAnnex.DeterminantFrame.abs_inverse_mulVec_apply_le`
- coordinate factor — `MathlibAnnex.DeterminantFrame.coordinateInverseFactor`

- ambient inverse constant — `MathlibAnnex.DeterminantFrame.inverseBoundConstant`
- positivity of that constant — `MathlibAnnex.DeterminantFrame.inverseBoundConstant_pos`
- the common inverse-bound structure — `MathlibAnnex.DeterminantFrame.NearMaxInverseBound`
- the nonzero determinant criterion — `MathlibAnnex.DeterminantFrame.nearMaxFrame_det_ne_zero`
- reconstruction through the inverse frame —
`MathlibAnnex.DeterminantFrame.inverse_mulVec_frameCoordinates`
- frame-map injectivity — `MathlibAnnex.DeterminantFrame.frameMap_injective_of_det_ne_zero`
- evaluation on the basis — `MathlibAnnex.DeterminantFrame.functionalCoordinates`
- expansion of a functional in these coordinates —
`MathlibAnnex.DeterminantFrame.functional_apply_eq_sum`
- the transposed Cramer replacement —
`MathlibAnnex.DeterminantFrame.replacementDeterminant_eq_cramerTranspose`
- row replacement operation — `MathlibAnnex.DeterminantFrame.replaceRow`
- replacement determinant — `MathlibAnnex.DeterminantFrame.replacementDeterminant`
- Exact declaration and proof — `MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`
- `frameMap` — `MathlibAnnex.DeterminantFrame.frameMap`
- `frameCoordinates` — `MathlibAnnex.DeterminantFrame.frameCoordinates`

Lean source signature (exact)

```

/-- The explicit Cramer estimate transported back through the inverse coordinate map. -/
theorem inverseBoundConstant_bound (b : Basis (Fin n) ℝ E)
  {η : ℝ} (hηD : η < determinantMaximum b) {B : Frame n E}
  (hB : B ∈ nearMaxFrames b η) (c : Fin n → ℝ) :
  ‖(coordinateEquiv b).symm ((frameMatrix b B)-1.mulVec c)‖ ≤
    inverseBoundConstant b η * ‖c‖

```

The hypotheses are `hηD` and `hB`; `c` is an arbitrary vector, not a further bound.

In the source	Mathematical meaning
<code>n; b : Basis (Fin n) ℝ E</code>	A nonnegative integer n and the ordered basis $(b_j)_{j=1}^n$ of E .
<code>[NormedAddCommGroup E] [NormedSpace ℝ E]</code> <code>[FiniteDimensional ℝ E]</code>	The ambient E is a finite-dimensional real normed vector space, with its given norm (not necessarily a coordinate sup norm). These hypotheses occur in the separate surrounding excerpt.
<code>Fin n → ℝ</code>	The coordinate space $\mathbb{R}^n$ with the sup norm. Lean indexes the coordinates by $0, \dots, n-1$; the formulas here use $1, \dots, n$.
<code>Frame n E</code>	The ordered families $F = (\ell_i)_{i=1}^n$ with $\ell_i : E \rightarrow \mathbb{R}$ continuous and real linear. No independence is part of this type.
<code>determinantMaximum b</code>	The real number $D_b = \max_{F \in \mathcal{U}} \det(\ell_i(b_j))_{i,j} $, using the specified basis b .

In the source	Mathematical meaning
<code>hηD : η < determinantMaximum b</code>	The input hypothesis $\eta < D_b$.
<code>hB : B ∈ nearMaxFrames b η</code>	The frame F called B satisfies $\ \ell_i\ _{E^*} \leq 1$ for every i and $D_b - \eta \leq \det M_F $. Together with the gap, this implies $\det M_F \neq 0$.
<code>(c : Fin n → ℝ); ‖c‖</code>	An arbitrary coordinate vector $c \in \mathbb{R}^n$ and its sup norm $\ c\ _\infty$.
<code>frameMatrix b B</code>	The real $n \times n$ matrix $M_F = (\ell_i(b_j))_{i,j}$.
<code>(frameMatrix b B)⁻¹.mulVec c</code>	The vector $M_F^{-1}c$. ⁻¹ is matrix inverse and mulVec is matrix multiplication by a column.
<code>(coordinateEquiv b).symm (...)</code>	The vector $C_b^{-1}(M_F^{-1}c) \in E$. symm selects the inverse coordinate isomorphism; the norm outside this expression is the norm of E .

In the source	Mathematical meaning
<code>inverseBoundConstant b η</code>	$K_b(\eta) = 1 + \ C_b^{-1}\ _{\text{op}} \frac{nD_b}{\alpha_b(D_b - \eta)}$, where $\alpha_b = (1 + \sum_i \ e_i^*\ _{E^*})^{-1}$ and $e_i^*(x) = (C_b x)_i$.
<code>‖(coordinateEquiv b).symm ((frameMatrix b B)⁻¹.mulVec c)‖ ≤ inverseBoundConstant b η * ‖c‖</code>	The complete conclusion $\ C_b^{-1}(M_F^{-1}c)\ _E \leq K_b(\eta)\ c\ _\infty$, for this frame and every supplied coefficient vector.

Braces allow Lean to infer a parameter. Parentheses provide explicit arguments. $\rightarrow \mathbb{L}[\mathbb{R}]$ in the surrounding definitions denotes a continuous real linear map.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 47–49:

```
variable {n : ℕ}
variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]
[FiniteDimensional ℝ E]
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 37–49. The full original context is retained with the source evidence.

▼ Exact content identity

Declaration: MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound

Accepted content SHA-256: 19f4ef4c81f82be19fbdddbef2b0612b3e7a770bdf8a166c617b562568245df9

Accepted source guide SHA-256: ebd0581a963aededaa6753c3d0ec9e22b754fde2d8f624d16a916fc2d5e495fc

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73